Theme/Title Bachelor Thesis AGREEMENT

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Specialization: Computer Science in English

Class academic years: 2022 - 2025

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Thesis domain: Music Recommendation Systems

Thesis title: The Development of a Mobile Context-Aware Music Recommendation System Application

Bachelor thesis topic description:

The AfterVibes project will be a mobile, Android exclusive, application that will improve the listening to music experience of the user in real time based on external factors. Unlike common music apps like Spotify, Apple Music or Youtube that base their recommendations on static preferences, AfterVibes will consider the activity level of the user (walking, running, sitting), the time of the day, mood check-ins and the weather outside. The goal of the project is to ensure that the music recommended and played aligns with the surroundings and the mood of the user.

To achieve this, for the music playback part, the application will use ExoPlayer for local streaming and Spotify integration for dynamic playlist generation. Users will be able to use algorithm-curated playlists, with playback controls directly implemented inside the application. For activity recognition the app will integrate the Google Health Connect API to monitor the user's movement and activity levels, by looking at metrics such as: heart rate and step count. For instance, soothing melodies may be recommended during periods of low activity, while energetic tracks will be suggested during more active moments. The application will also take into consideration the time of day and the weather conditions, retrieved from OpenWeatherMap API, to further refine recommendations.

The project will be structured in the following way:

- Backend: Python, Firebase and FastAPI – where the actual recommendation algorithm will be implemented, manages user data.
- Integration Layer: used to connect the app to Spotify for playback, ExoPlayer for local streaming, Health Connect for tracking activity and OpenWeatherMap for weather awareness

- Fronted: React Native with Expo – user interactions

By combining the seamless music playback and real-time activity with context-aware recommendation algorithm, the application will offer a new approach to music selection, constantly adapting based on the user's lifestyle. The research aspect of this thesis will explore real-time user activity detection, optimizing the playback experience and how music can be categorized based on the factors mentioned above.

Date, 03.03.2025 **Student signature,** 

Scientific coordinator signature, 